

Machine Learning Fundamentals - Study Notes

Original study notes prepared for the DataMind RAG Assistant project.

What is machine learning?

Machine learning is a set of techniques that let computer programs improve their performance on a task by learning patterns from data, rather than being explicitly programmed with fixed rules for every situation.

It is commonly split into supervised learning (learning from labeled examples), unsupervised learning (finding structure in unlabeled data), and reinforcement learning (learning by trial and error through rewards and penalties).

Supervised learning: regression and classification

In supervised learning, a model is trained on input-output pairs so it can predict the output for new, unseen inputs. Regression tasks predict a continuous number, such as a house price, while classification tasks predict a discrete category, such as whether an email is spam.

Common supervised algorithms include linear and logistic regression, decision trees, random forests, gradient-boosted trees, and neural networks — each with different trade-offs in interpretability, training speed, and predictive performance.

Feature engineering

Feature engineering is the process of transforming raw data into informative input variables (features) that make it easier for a model to find useful patterns. This can include creating new columns (like extracting the day-of-week from a date), encoding categorical variables as numbers, and scaling numeric features to comparable ranges.

Good feature engineering often has a bigger impact on model performance than the choice of algorithm itself, because it directly shapes what information the model has access to.

Train/test split and overfitting

A train/test split divides the available data into a portion used to fit the model (the training set) and a portion held back purely for evaluation (the test set), so that performance is measured on data the model has never seen during training.

Overfitting happens when a model learns the noise and idiosyncrasies of the training data too closely, achieving excellent training performance but generalizing poorly to new data.

Underfitting is the opposite problem: the model is too simple to capture the real patterns, so it performs poorly on both training and test data.

Cross-validation extends this idea by splitting the data into several folds and rotating which fold is held out for testing, giving a more reliable estimate of how a model will perform on new data than a single train/test split.

Model evaluation

For regression, common evaluation metrics include mean absolute error (MAE), mean squared error (MSE), and R-squared, which measures how much of the variance in the target the model explains.

For classification, common metrics include accuracy, precision, recall, and F1-score. Accuracy alone can be misleading on imbalanced datasets (for example, if 95% of emails are not spam, a model that always predicts 'not spam' gets 95% accuracy while being useless), which is why precision and recall are often reported alongside it.